SCUDEM XI, Problem B: model notes

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Note: I had Claude help with the LaTeX and the diagrams. I figured this would be more readable than my handwritten notes.

Two spider monkeys, A and B, forage in the same forest and can tip each other off about food. State = what mode each monkey is in, not where it is. Modes: **S** searching, **T** traveling to a tip, $\mathbf{E}_a$ eating alone, $\mathbf{E}_s$ eating a shared patch (only together). Every arrow is a rate; the probability of each pair state obeys $\frac{d}{dt}P =$ flow in $-$ flow out.

1 Parameters

Chosen

	meaning	units	default
environment			
A	forest area	L^2	100^2
n	number of good patches	count	13
w	patch width (gaussian σ)	L	3
amp	peak food density of a fresh patch	food/L^2	10
m	density a monkey notices	food/L^2	0.5
κ	rate a patch vanishes on its own (rot, fall, competitors)	T^{-1}	0.01
monkey			
v	travel speed	L/T	0.5
r	vision radius	L	2
c	eat rate	T^{-1}	0.03
d	separation of the pair	L	10 (swept)
communication			
C	channel capacity	bits/T	10
b	bits per tip	bits	10
σ_0	tip blur per unit distance	none (rad)	0.05
R_c	contact range	L	30
p_A, p_B	accuracy of tips A, B send $(1 - p_{\text{lie}})$	prob.	1
ϕ	forgiveness rate (a feud lasts $1/\phi$)	T^{-1}	1/1500

Derived

	formula	meaning	units	at defaults
λ	$2rv\,n/A$	discovery rate while searching	T^{-1}	0.0026
μ	$c/\ln(\text{amp}/m)$	patch depletion rate, one eater	T^{-1}	0.0100
F	$2\pi\,\text{amp}\,w^2$	food in a fresh patch	food	565
e	μF	intake rate while eating	food/T	5.66
$c(d)$	1 if $d \le R_c$ else 0	contact factor	none	1
$o(d)$	$\frac{1}{n} + \left(1 - \frac{1}{n}\right) e^{-d/(r+w)}$	overlap: chance a find is in shared ground	none	0.20
a_ℓ	$\max\left(A\,2^{-b},\ \pi(\sigma_0 d)^2\right)$	area left to search after a tip	L^2	9.77
τ	$d/v + a_\ell/(2rv)$	mean trip time	T	24.9
ν	C/b	tip rate	T^{-1}	1

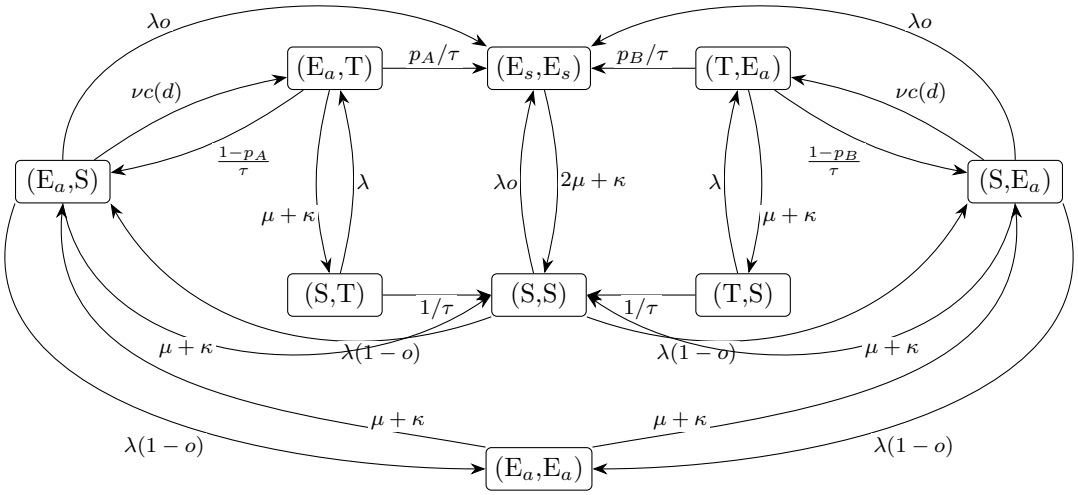
Notes: a patch with one eater ends at rate $\mu + \kappa$ (eaten out or vanishes); a shared patch at $2\mu + \kappa$. μ solves $\text{amp}\,e^{-ct} = m$ for the patch lifetime. F is the polar integral of the gaussian. λ is the swept strip $2r$ wide. Each bit halves the area left to search, but never below the blur circle.

2 The 9-state model

States: (S,S), (S,E$_a$), (E$_a$,S), (E$_a$,E$_a$), (T,E$_a$), (E$_a$,T), (T,S), (S,T), (E$_s$,E$_s$), written (A's mode, B's mode).

A's arrows (B mirrors, with $p_A \leftrightarrow p_B$)

from	to	rate	
(S,S)	(E$_a$,S)	$\lambda(1-o)$	A finds its own patch
(S,S)	(E$_s$,E$_s$)	λo	find in shared ground, both eat (one arrow)
(S,E$_a$)	(E$_a$,E$_a$)	$\lambda(1-o)$	A finds its own patch
(S,E$_a$)	(E$_s$,E$_s$)	λo	A stumbles onto B's patch
(S,E$_a$)	(T,E$_a$)	$\nu c(d)$	B tips A
(T,E$_a$)	(E$_s$,E$_s$)	p_B/τ	true tip, they share
(T,E$_a$)	(S,E$_a$)	$(1-p_B)/\tau$	false tip, nothing there
(T,E$_a$)	(T,S)	$(\mu+\kappa)$	B's patch ends while A is en route
(T,S)	(S,S)	$1/\tau$	A arrives, patch gone
(T,S)	(T,E$_a$)	λ	B finds a new patch and re-tips
(E$_a$,S)	(S,S)	$(\mu+\kappa)$	A's patch ends
(E$_a$,E$_a$)	(S,E$_a$)	$(\mu+\kappa)$	A's patch ends
(E$_s$,E$_s$)	(S,S)	$(2\mu+\kappa)$	shared patch ends (one arrow)



Equations

P_{XY} is the probability of state (X,Y): first letter is A's mode, second is B's, and E means E_a (so $P_{SE} = P(\mathrm{S}, \mathrm{E}_a)$). $P_{E_sE_s}$ is the shared state.

$$
\begin{aligned}
\dot{P}_{SS} &= (\mu+\kappa)(P_{SE}+P_{ES}) + (2\mu+\kappa)P_{E_sE_s} + \tfrac{1}{\tau}(P_{ST}+P_{TS}) - \lambda(2-o)\,P_{SS} \\
\dot{P}_{SE} &= \lambda(1-o)P_{SS} + (\mu+\kappa)P_{EE} + \tfrac{1-p_B}{\tau}P_{TE} - (\lambda+\nu c(d)+\mu+\kappa)P_{SE} \\
\dot{P}_{ES} &= \lambda(1-o)P_{SS} + (\mu+\kappa)P_{EE} + \tfrac{1-p_A}{\tau}P_{ET} - (\lambda+\nu c(d)+\mu+\kappa)P_{ES} \\
\dot{P}_{EE} &= \lambda(1-o)(P_{SE}+P_{ES}) - 2(\mu+\kappa)P_{EE} \\
\dot{P}_{TE} &= \nu c(d)\,P_{SE} + \lambda P_{TS} - (\tfrac{1}{\tau}+\mu+\kappa)P_{TE} \\
\dot{P}_{ET} &= \nu c(d)\,P_{ES} + \lambda P_{ST} - (\tfrac{1}{\tau}+\mu+\kappa)P_{ET} \\
\dot{P}_{TS} &= (\mu+\kappa)P_{TE} - (\tfrac{1}{\tau}+\lambda)P_{TS} \\
\dot{P}_{ST} &= (\mu+\kappa)P_{ET} - (\tfrac{1}{\tau}+\lambda)P_{ST} \\
\dot{P}_{E_sE_s} &= \lambda o\,(P_{SS}+P_{SE}+P_{ES}) + \tfrac{p_B}{\tau}P_{TE} + \tfrac{p_A}{\tau}P_{ET} - (2\mu+\kappa)P_{E_sE_s}
\end{aligned}
$$

This is linear, $\dot{P} = PQ$, and the steady state solves $PQ = 0$ with $\sum P = 1$.

Readout. Intake of A $= e \cdot P(\text{A eating}) = e\,(P_{ES} + P_{EE} + P_{ET} + P_{E_sE_s})$; B mirrors. **Silent** (no communication) = same chain with channel capacity $C = 0$, so the tip rate $\nu = C/b = 0$ and no T state is ever reached. (Careful: C is channel capacity, c is eat rate, and $c(d)$ is the contact factor.)

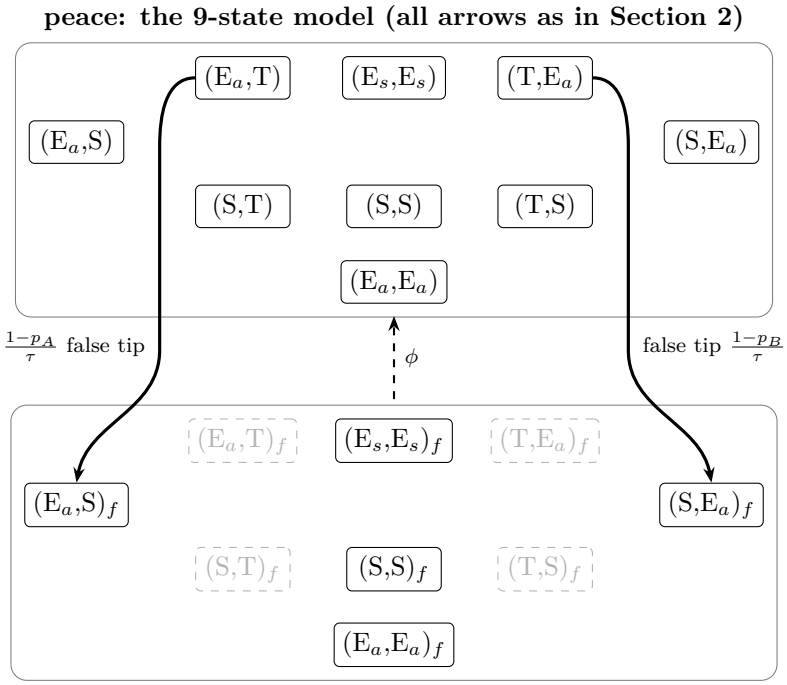
3 The 14-state model: adding a feud

This model is best understood as an extension of the 9-state model in Section 2, which is why both are included: everything in Section 2 still holds, and the feud adds one mechanism on top.

A false tip (A arrives, nothing there, B still eating) starts a **feud**: no tips until forgiven. The feud is a copy of the 9-state diagram with the tip arrows $\nu c(d)$ set to 0. The T boxes become unreachable, leaving 5 boxes; this is just the silent model. Two changes connect the copies:

- the false-tip arrow $(T, E_a) \xrightarrow{(1-p_B)/\tau} (S, E_a)$ now lands in the feud copy $(S, E_a)_f$ (mirror for A);
- every feud box returns to its twin at rate ϕ.

The receiver can't tell a lie from an honest mistake, so both start feuds.



Changes to the equations

In the 9 peace equations, replace $\frac{1-p_B}{\tau}P_{TE}$ in $\dot{P}_{SE}$ with $\phi P_{SE,f}$ (and $\frac{1-p_A}{\tau}P_{ET}$ in $\dot{P}_{ES}$ with $\phi P_{ES,f}$), and add $+\phi P_{X,f}$ to $\dot{P}_{SS}$, $\dot{P}_{EE}$, $\dot{P}_{E_sE_s}$. The feud copy:

$$\begin{aligned}
\dot{P}_{SS,f} &= (\mu+\kappa)(P_{SE,f}+P_{ES,f}) + (2\mu+\kappa)P_{E_sE_s,f} - (\lambda(2-o)+\phi)P_{SS,f}\\
\dot{P}_{SE,f} &= \lambda(1-o)P_{SS,f} + (\mu+\kappa)P_{EE,f} + \tfrac{1-p_B}{\tau}P_{TE} - (\lambda+\mu+\kappa+\phi)P_{SE,f}\\
\dot{P}_{ES,f} &= \lambda(1-o)P_{SS,f} + (\mu+\kappa)P_{EE,f} + \tfrac{1-p_A}{\tau}P_{ET} - (\lambda+\mu+\kappa+\phi)P_{ES,f}\\
\dot{P}_{EE,f} &= \lambda(1-o)(P_{SE,f}+P_{ES,f}) - (2\mu+2\kappa+\phi)P_{EE,f}\\
\dot{P}_{E_sE_s,f} &= \lambda o\,(P_{SS,f}+P_{SE,f}+P_{ES,f}) - (2\mu+\kappa+\phi)P_{E_sE_s,f}
\end{aligned}$$

Intake counts eating in both copies. As $\phi \to \infty$ (forgive instantly) this is exactly the 9-state model.

Assumptions to state: all waits are exponential (a trip takes *mean* τ); the environment is constant (n fixed, since unclaimed patches turn over at rate κ); the receiver always follows a tip outside a feud.